

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 18)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1	.	1
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Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

0 1 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

0	1	.	3
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Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

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Water flow

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Day length

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Temperature

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10

02 . 1 Define each of the following ecological terms and give an example.

[8 marks]

Species

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Ecological niche

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Community

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Biome

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0 2 . 2

Explain the difference between an ecosystem and a biome.

[2 marks]

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10

0 3

Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 3 . 1

Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer

 %

0 3 . 2

Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 3 . 3

Explain why few food chains have more than four trophic levels.

[2 marks]

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0 3 . 4

Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

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0 3 . 5

Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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10

0 4

Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 4 . 1

Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

0	4	.	2
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Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

2

0	4	.	3
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Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

10

05 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

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DNA databases and eDNA

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Image recognition software

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Acoustic monitoring / sonograms

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05 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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06

Table 1 shows the number of plant species per m² in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m ²	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

06

1

Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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06

2

Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

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06

3

Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

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06 . 4

Explain how burning maintains heathland.

[2 marks]

10

07 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

07 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

07 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

07 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

10

08 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

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Fuel

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Food

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Fibres

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Medicines

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08 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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09 . 1

Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

- A A sample taken from the most convenient location ☐
- B A sub-sample that accurately reflects the complete data set ☐
- C The largest sample that can be collected ☐
- D A sample that supports the hypothesis ☐

09 . 2

Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

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09 . 3

Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

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10 . 1

Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Do not write outside the box

Extra space

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10 . **2** State what is meant by **permaculture**.

[1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

1 1 . 2

Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS